

LAB 4 – Fuzzy Inference System

Membership Functions & IF-THEN Rules

Cluster A (cluster_id = 2, n = 28 subjects) | Cluster B (cluster_id = 3, n = 51 subjects)

Dataset: knn_global | SOM: 12×12 | Linkage: complete, cityblock

Cluster A: 6 features, 7 centroid rules + 76 pairwise (83 unique) | Cluster B: 4 features, 8 centroid rules + 46 pairwise (54 unique)

Color Legend

Fuzzy Term Colors (Antecedent)

LOW / STABLE / ABSENT

Feature value is low or condition is absent
→ generally associated with lower risk

MEDIUM / VARIABLE / NORMAL

Feature value is in the medium range
→ intermediate or uncertain condition

HIGH / UNSTABLE / PRESENT

Feature value is high or condition is present
→ generally associated with higher risk

— (not in rule)

This feature is not part of the rule antecedent

Consequent Colors (Clinical Risk)

LOW (Success)

Score < Low/Medium threshold
→ predicted extubation success

MEDIUM (Failure)

Score between thresholds
→ predicted extubation failure

HIGH (Death)

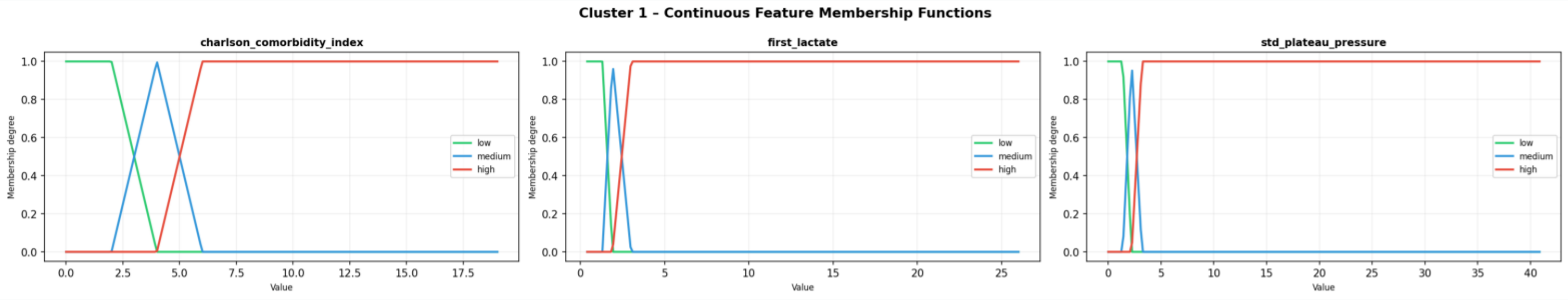
Score > Medium/High threshold
→ predicted death during ICU

Rules are derived from hierarchical clustering (complete linkage, cityblock) applied per outcome class.
Centroid rules (full-feature) are generated from cluster centroids. Pairwise rules extend coverage using all pairs of features.

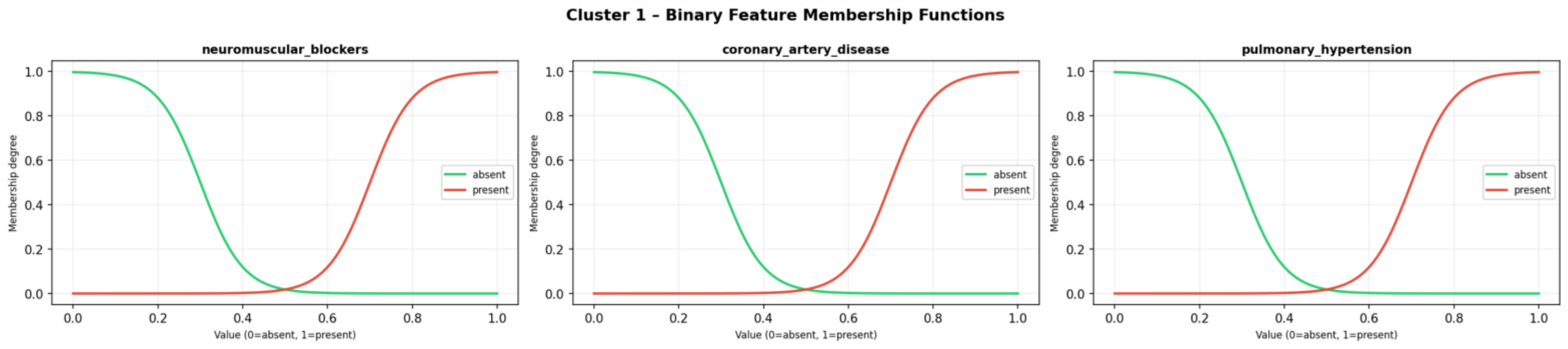
Cluster A

cluster_id = 2 | n = 28 subjects | 6 features

Cluster A - Membership Functions (Continuous Features)



Cluster A - Membership Functions (Binary Features)



Cluster A - Centroid IF-THEN Rules (7 rules, full-feature)

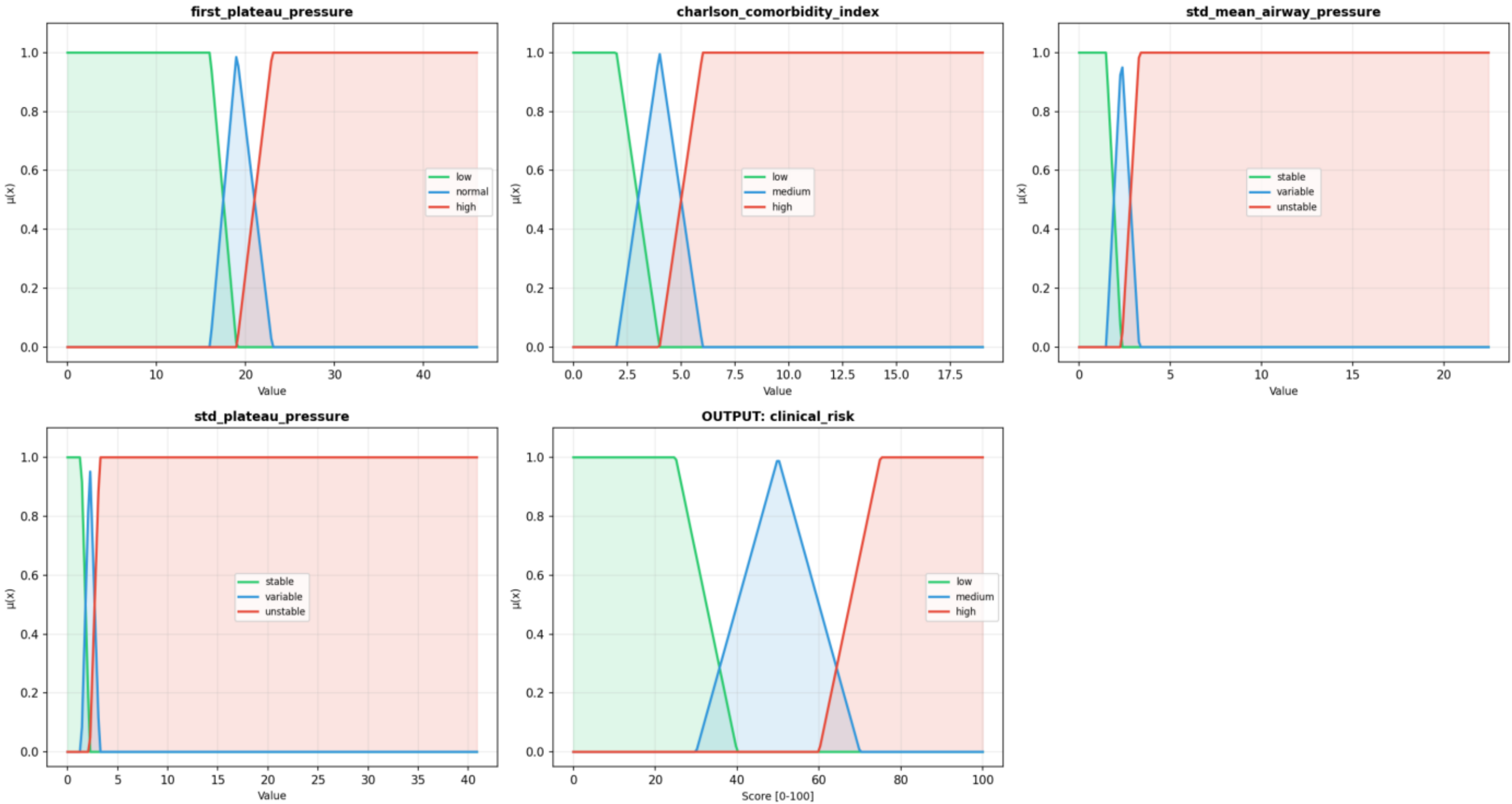
#	Neuromusc. Blockers	Coronary Art. Dis.	Charlson Index	First Lactate	Pulm. Hypert.	Std Plateau Pressure		Clinical Risk
1	ABSENT	ABSENT	MEDIUM	LOW	ABSENT	MEDIUM	→	MEDIUM (Failure)
2	ABSENT	PRESENT	HIGH	LOW	ABSENT	MEDIUM	→	LOW (Success)
3	ABSENT	ABSENT	HIGH	LOW	ABSENT	HIGH	→	LOW (Success)
4	ABSENT	ABSENT	MEDIUM	HIGH	ABSENT	LOW	→	LOW (Success)
5	ABSENT	ABSENT	LOW	HIGH	ABSENT	MEDIUM	→	HIGH (Death)
6	ABSENT	ABSENT	HIGH	HIGH	ABSENT	MEDIUM	→	HIGH (Death)
7	ABSENT	PRESENT	HIGH	HIGH	ABSENT	MEDIUM	→	HIGH (Death)

Cluster B

cluster_id = 3 | n = 51 subjects | 4 features

Cluster B - Membership Functions (Continuous Features)

LAB4 - Cluster 4: Membership Functions (Feature Continue + Output)



Cluster B - Centroid IF-THEN Rules (8 rules, full-feature)

#	First Plateau Pressure	Std Mean Airway Pressure	Charlson Index	Std Plateau Pressure		Clinical Risk
1	NORMAL	VARIABLE	LOW	UNSTABLE	→	MEDIUM (Failure)
2	LOW	VARIABLE	LOW	VARIABLE	→	MEDIUM (Failure)
3	HIGH	UNSTABLE	MEDIUM	UNSTABLE	→	LOW (Success)
4	LOW	STABLE	HIGH	STABLE	→	LOW (Success)
5	LOW	VARIABLE	MEDIUM	VARIABLE	→	LOW (Success)
6	LOW	VARIABLE	HIGH	VARIABLE	→	HIGH (Death)
7	HIGH	VARIABLE	HIGH	UNSTABLE	→	HIGH (Death)
8	NORMAL	UNSTABLE	LOW	UNSTABLE	→	HIGH (Death)

Cluster B - Pairwise IF-THEN Rules (46 rules, 2-feature pairs) (page 1)

#	First Plateau Pressure	Std Mean Airway Pressure	Charlson Index	Std Plateau Pressure		Clinical Risk
9	NORMAL	VARIABLE	—	—	→	MEDIUM (Failure)
10	NORMAL	—	LOW	—	→	MEDIUM (Failure)
11	NORMAL	—	—	UNSTABLE	→	MEDIUM (Failure)
12	—	VARIABLE	LOW	—	→	MEDIUM (Failure)
13	—	VARIABLE	—	UNSTABLE	→	MEDIUM (Failure)
14	—	—	LOW	UNSTABLE	→	MEDIUM (Failure)
15	LOW	VARIABLE	—	—	→	MEDIUM (Failure)
16	LOW	—	LOW	—	→	MEDIUM (Failure)
17	LOW	—	—	VARIABLE	→	MEDIUM (Failure)
18	—	VARIABLE	—	VARIABLE	→	MEDIUM (Failure)
19	—	—	LOW	VARIABLE	→	MEDIUM (Failure)
20	HIGH	UNSTABLE	—	—	→	LOW (Success)
21	HIGH	—	MEDIUM	—	→	LOW (Success)
22	HIGH	—	—	UNSTABLE	→	LOW (Success)

Cluster B - Pairwise IF-THEN Rules (46 rules, 2-feature pairs) (page 2)

#	First Plateau Pressure	Std Mean Airway Pressure	Charlson Index	Std Plateau Pressure		Clinical Risk
23	—	UNSTABLE	MEDIUM	—	→	LOW (Success)
24	—	UNSTABLE	—	UNSTABLE	→	LOW (Success)
25	—	—	MEDIUM	UNSTABLE	→	LOW (Success)
26	LOW	STABLE	—	—	→	LOW (Success)
27	LOW	—	HIGH	—	→	LOW (Success)
28	LOW	—	—	STABLE	→	LOW (Success)
29	—	STABLE	HIGH	—	→	LOW (Success)
30	—	STABLE	—	STABLE	→	LOW (Success)
31	—	—	HIGH	STABLE	→	LOW (Success)
32	LOW	VARIABLE	—	—	→	LOW (Success)
33	LOW	—	MEDIUM	—	→	LOW (Success)
34	LOW	—	—	VARIABLE	→	LOW (Success)
35	—	VARIABLE	MEDIUM	—	→	LOW (Success)
36	—	VARIABLE	—	VARIABLE	→	LOW (Success)

Cluster B - Pairwise IF-THEN Rules (46 rules, 2-feature pairs) (page 3)

#	First Plateau Pressure	Std Mean Airway Pressure	Charlson Index	Std Plateau Pressure		Clinical Risk
37	—	—	MEDIUM	VARIABLE	→	LOW (Success)
38	LOW	VARIABLE	—	—	→	HIGH (Death)
39	LOW	—	HIGH	—	→	HIGH (Death)
40	LOW	—	—	VARIABLE	→	HIGH (Death)
41	—	VARIABLE	HIGH	—	→	HIGH (Death)
42	—	VARIABLE	—	VARIABLE	→	HIGH (Death)
43	—	—	HIGH	VARIABLE	→	HIGH (Death)
44	HIGH	VARIABLE	—	—	→	HIGH (Death)
45	HIGH	—	HIGH	—	→	HIGH (Death)
46	HIGH	—	—	UNSTABLE	→	HIGH (Death)
47	—	VARIABLE	—	UNSTABLE	→	HIGH (Death)
48	—	—	HIGH	UNSTABLE	→	HIGH (Death)
49	NORMAL	UNSTABLE	—	—	→	HIGH (Death)
50	NORMAL	—	LOW	—	→	HIGH (Death)

Cluster B - Pairwise IF-THEN Rules (46 rules, 2-feature pairs) (page 4)

#	First Plateau Pressure	Std Mean Airway Pressure	Charlson Index	Std Plateau Pressure		Clinical Risk
51	NORMAL	—	—	UNSTABLE	→	HIGH (Death)
52	—	UNSTABLE	LOW	—	→	HIGH (Death)
53	—	UNSTABLE	—	UNSTABLE	→	HIGH (Death)
54	—	—	LOW	UNSTABLE	→	HIGH (Death)